

The tourism industry is becoming increasingly reliant on AI, as the latter can help with a variety of travel-related tasks including booking hotels, flights, and the best routes for consumers. For better and faster customer service, chatbots driven by artificial intelligence are being used in the travel industry.

Machine learning in different domains

All techniques and tools that let software applications and gadgets respond and develop on their own are referred to as machine learning (ML). AI can learn without really being explicitly programmed to perform the required action, thanks to machine learning techniques. Rather than relying on predefined computer instructions, the ML algorithm learns a pattern from sample inputs, and then anticipates and executes tasks completely based on the learned pattern. If rigorous algorithms aren't an option, machine learning can be a life-saver. It will pick up the new procedure by analysing prior ones and then putting it into action. ML has cleared the way for technical advancements and technologies that were previously unimaginable in a variety of industries. It is used in a variety of cutting-edge technologies today — from predictive algorithms to Internet TV live streaming.

The way our email service providers assist us in dealing with spam is one example of a machine learning application. Many e-commerce organisations use ML techniques in conjunction with other IT security solutions to prevent forgery and improve the efficiency of their recommendation engine.

A notable ML and AI technique is image recognition, which is a method for categorising and detecting a feature or an item in a digital image. Classification and face recognition are done using this method.

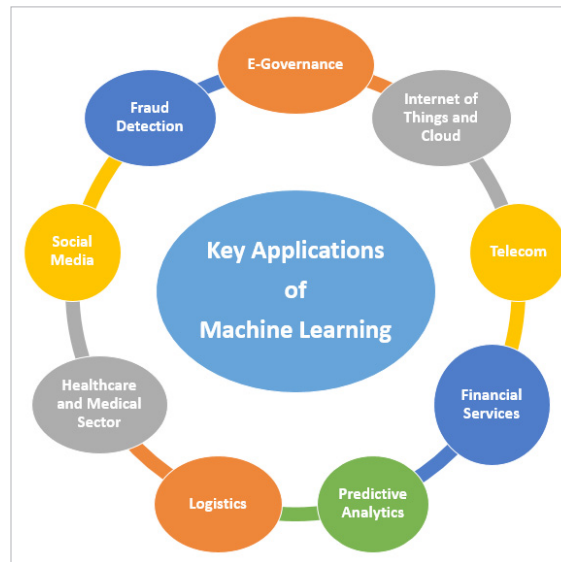


Figure 1: Key applications of machine learning

The use of machine learning for recommender systems is among its

most widely used and well-known applications. In today's e-commerce world, product recommendation is a prominent tool that utilises powerful machine learning techniques. Websites use AI and ML to keep track of past purchases, search trends, and shopping cart history, and then generate product recommendations based on that data.

Machine learning's most critical

application is sentiment analysis. It's a real-time machine learning

Table 1: Tools and frameworks for machine learning

Tool/Platform	URL
Streamlit	https://github.com/streamlit/streamlit
TensorFlow	https://www.tensorflow.org/
PyTorch	https://pytorch.org/
scikit-learn	https://scikit-learn.org/
Apache Spark	https://spark.apache.org/
Torch	http://torch.ch/
Hugging Face	https://huggingface.co/
Keras	https://keras.io/
TensorFlowJS	https://www.tensorflow.org/js
KNIME	https://www.knime.com/
Apache Mahout	https://mahout.apache.org/
Accord	http://accord-framework.net/
Shogun	http://shogun-toolbox.org/
RapidMiner	https://rapidminer.com/
Blocks	https://github.com/mila-ijia/blocks
TuriCreate	https://github.com/apple/turicreate
Dopamine	https://github.com/google/dopamine
FlairNLP	https://github.com/flairNLP/flair